Shivam Maurya

Computer Vision Engineer | Deep Learning, Real-Time Perception and Edge AI

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SUMMARY

Computer Vision Engineer (B.E., CGPA 8.06 / 10) who builds and deploys real-time perception systems: object detection (YOLOv8), semantic segmentation (U-Net), and transfer-learning classifiers on medical imaging (up to 99.25% accuracy). Engineered a ROS 2 vision pipeline running at 257 FPS on CPU with ONNX Runtime and hot-swappable model backends, and shipped edge CV dashboards with live MJPEG streaming and GPS tagging. Experienced in model benchmarking, explainability (Grad-CAM), and synthetic data generation with calibrated RTX camera arrays in NVIDIA Isaac Sim. All India Rank under 20, ABU Robocon; Amazon ML Challenge 2026 Top 500; 15+ merged open-source pull requests.

TECHNICAL SKILLS

Computer Vision: Object Detection (YOLOv8, YOLOv11), Semantic Segmentation (U-Net), Image Classification, Transfer Learning (Xception, ResNet-18, MobileNetV2, EfficientNetB0), Data Augmentation, OCR (Tesseract), Real-Time Video Processing, MJPEG Streaming, Camera Calibration, Vision-Based Robot Alignment

Deep Learning: PyTorch, TensorFlow, Keras, CNN, ONNX, ONNX Runtime, TensorRT Inference, Model Benchmarking, Hyperparameter Tuning, Cross-Validation, Grad-CAM, SHAP, LIME

Libraries and Languages: Python, C++, OpenCV, NumPy, Pandas, scikit-learn, SQL, Bash

Robotics and Simulation: ROS 2 (Humble, Jazzy), MoveIt 2, Nav2, NVIDIA Isaac Sim (RTX camera arrays, synthetic data generation), Gazebo, RViz2

Edge AI and Deployment: Edge AI, CPU Inference Optimization, Flask, FastAPI, WebSockets, REST APIs, Docker, Git, GitHub, CI/CD, Linux, GPS / NMEA Integration

Foundations: Probability and Statistics, Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems

PROJECTS

RealFake Classifier: Real-Time Perception Pipeline for ABU Robocon | ONNX Runtime, YOLOv8, TensorFlow, ROS 2, Flask | github.com/ShivamMaurya14/RealFake-Classifier-Robocon

- Engineered a CPU-only ROS 2 perception pipeline with hot-swappable ONNX, YOLOv8, and TensorFlow CNN backends that switch at runtime without node restarts.
- Benchmarked backends on identical input: ONNX Runtime 3.89 ms (257 FPS), YOLOv8 about 18.5 ms (about 54 FPS), TensorFlow 22.18 ms (45.1 FPS); shipped the fastest as the default.
- Built a decision engine that maps bounding boxes to gripper commands (PICK_REAL, REJECT_FAKE) and alignment targets (optical-center offset dx/dy, heading error), plus a Flask dashboard with MJPEG streaming and live telemetry.

Road Defect Detector: Edge Computer Vision Platform | YOLOv8, U-Net, OpenCV, TensorFlow, Flask | github.com/ShivamMaurya14/Road-Defect-Detector

- Built a 3-mode defect detection platform (raw GPS survey, U-Net semantic segmentation of cracks, YOLOv8 bounding-box detection of potholes) deployable on dashcams, survey trucks, and inspection robots.
- Integrated a GPS telemetry pipeline that burns live coordinates and vehicle speed into MP4 recordings, supporting browser geolocation, NMEA serial GPS (NEO-6M/8M), and mock GPS simulation.
- Designed a Flask dashboard with MJPEG live streaming, a Leaflet map with geotagged defect pins, and CSV export for log management.

MedSynapse: Medical Imaging AI with 9 Diagnostic Modules | TensorFlow, PyTorch, Xception, U-Net, FastAPI, React | IIIT Pune Finalist, Avishkar 2026 | github.com/ShivamMaurya14/MedSynapse

- Trained 5 imaging models: pneumonia on chest X-rays (Xception, 83.01%), brain tumor on MRI (Xception, 95.25%), kidney pathology on CT (MobileNetV2 + EfficientNetB0 + U-Net, 99.25%), skin cancer (hierarchical CNN, 97.98%), and eye disease on fundus images (ResNet-18, 90.78%).
- Added Grad-CAM, SHAP, and LIME explainability with SHAP-LIME agreement scoring, and a multimodal ingestion pipeline for X-rays, MRI, fundus images, and lab PDFs using Tesseract OCR and PyMuPDF.

PCB Fault Detector: Automated Optical Inspection | YOLO, OpenCV, TensorRT, FastAPI | github.com/ShivamMaurya14/PCB-Fault-Detector

- Built a plug-and-play PCB defect classifier using YOLO classification (.pt and TensorRT .engine models) with browser webcam streaming, QR code traceability, and automated CSV inspection logging.

Multi-Arm UR10e Pick-and-Place with NVIDIA Isaac Sim | ROS 2 Jazzy, MoveIt 2, Isaac Sim, Gazebo | github.com/ShivamMaurya14/MultiArm-Pick-and-Place

- Built a procedural multi-cell spawner creating 30 UR10e robots and 20 RTX pinhole cameras running at 40-50 FPS on a single consumer GPU (RTX 5060 Ti 16GB).

- Integrated a calibrated camera array and tactile sensing for vision and tactile dataset generation for physical AI foundation models.

EXPERIENCE

Physical AI Intern | Nextup Robotics | Remote | Aug 2026 - Oct 2026

- Built a synchronized multimodal data-generation pipeline (20 calibrated RTX cameras plus tactile contact sensors across 7 rigid bodies per robot) producing vision and tactile datasets for NVIDIA Cosmos foundation models.
- Scaled the simulation to 30 UR10e robots at 40-50 FPS through profiling and buffer tuning, and authored 7 engineering guides covering setup, calibration, and headless CI deployment.

Software Lead | Centre of Excellence for AI & Robotics, Army Institute of Technology, Pune | Jun 2024 - Present

- Led software for the ABU Robocon national robot (All India Rank under 20), deploying real-time CNN perception into navigation and gripper decision loops at 30+ FPS on CPU-only hardware.
- Built an ONNX export pipeline and a JSON-configurable inference backend enabling runtime model swaps and A/B testing with zero downstream code changes.
- Trained and validated 4+ ML architectures (CNN, Xception transfer learning, Random Forest, XGBoost) with stratified cross-validation, data augmentation, and hyperparameter sweeps; directed a 100+ member research cell.

Open-Source Contributor | ROS 2 Ecosystem (navigation2, ros2_control, ros2_controllers) | Jun 2025 - Present

- Authored 15+ merged pull requests; diagnosed a controller crash from backtraces, fixed cross-repository CI failures, and added static type checking.

EDUCATION

Bachelor of Engineering (B.E.), Automation and Robotics Engineering | Army Institute of Technology, Pune, India | Jun 2024 - May 2028 (Expected) | **CGPA: 8.06 / 10**

Relevant Coursework: Computer Vision, Deep Learning, Machine Learning, Probability and Statistics, Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming

HONORS & ACHIEVEMENTS

- All India Rank under 20, ABU Robocon 2025/26 (Lead Software Architect)
- Top 500 All India, Amazon ML Challenge 2026 (F0.5 score 0.9602)
- State Level Round 2, Avishkar Maharashtra Inter-University Research Convention 2026 (MedSynapse)
- Finalist, IIIT Pune Thinking Machines 2025 (MedSynapse) | Runner-Up, Hackathon 2026 | Top Submission, CODECURE 2025, IIT BHU

Languages: English (Professional Working Proficiency), Hindi (Native)